Revision History

Date	Version	Description
October 11, 2019	0.50	Initial version created.
December 21, 2020	0.51	Updated various sections.
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Overview

The Atmosic ATM2/ATM3 series extreme low power Bluetooth 5.0 system-on-a-chip (SoC) incorporates an RF wakeup receiver, which is designed to decode an incoming RF paging signal to wake up the system. This document describes the functionality of the wakeup receiver, supported wakeup patterns, and reference applications. Figure 1 below conceptually shows the overall system.



Figure 1 - Conceptual System Block Diagram with Wakeup Receiver.

The Transmitter can be any of the following:

1. A hub device based on a Bluetooth Low Energy (LE) SoC, such as the Atmosic ATM2/ATM3. Access to the lower levels of the software stack is required to configure the SoC to transmit high duty cycle advertisements.
2. Android or other mobile devices that provide access for applications to run Bluetooth inquiry
3. iOS mobile devices that can send iBeacons
4. Any software-defined/programmable radio device that can send RF signals in a configurable on-off pattern. The transmitted RF energy is treated as an OOK (On-Off Keying) pattern and can be at any regulatory-compliant frequency such as the 2.4 GHz (Bluetooth/WiFi) or 900 MHz ISM bands

The Receiver is a device based on Atmosic ATM2/ATM3 SoC that includes an RF wakeup receiver antenna and match. Depending on the usage and availability of the Transmitter, different wakeup scenarios can be supported.

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RF Wakeup Receiver Functional Description

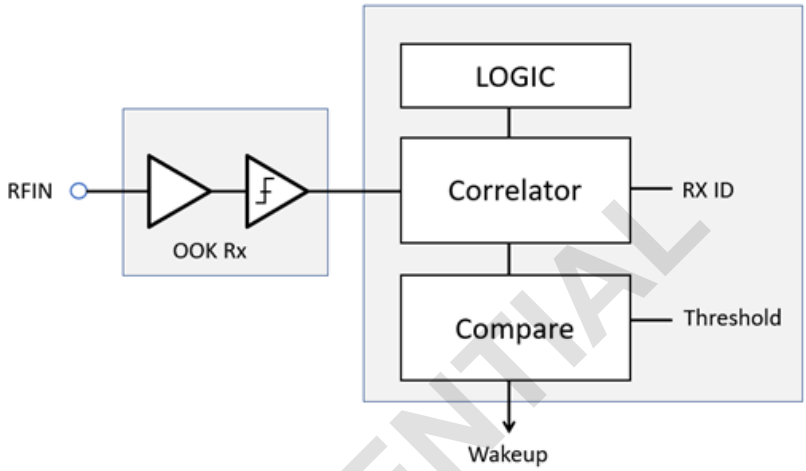


Figure 2 - Wakeup Receiver Front End Block Diagram

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The RF wakeup receiver consists of a dedicated RF input, analog front-end, and digital wakeup logic. When enabled, it continuously monitors the RF input for On-Off Keying (OOK) radio signals with ultra-low power consumption. The analog front-end is an active RF detector followed by amplification and quantization. The front-end output is processed by the digital wakeup logic based on programmable symbol parameters. When the incoming symbol sequence correlates and matches with an expected ID code, a wakeup event is declared. The detection threshold of the front end is continuously adjusted to provide robust operation even in noisy RF environments.

On the Evaluation Board (EVB), the wakeup receiver match is tuned to operate in the 2.4 GHz ISM band so that it can respond to RF paging signals based on Bluetooth packets. For applications with a Transmitter at other frequencies (such as the 900 MHz ISM band), please contact Atmosic Support for guidance on tuning the match.

Supported Wakeup Patterns

The SDK supports three fixed wakeup patterns via the REF_BCN environment variable:

1. Bluetooth LE high duty cycle connectable directed advertising
   Waveform details: 176 µs transmission at 1 ms interval
   REF_BCN setting: WURX_ADV
   Transmitter: EVB running BLE_adv_scan SDK example

2. Bluetooth inquiry
   Waveform details: two 68 µs transmissions 312.5 µs apart at 1.25 ms interval
   REF_BCN setting: WURX_PHONE_ADV
   Transmitter: Android device running the Atmosic app or any mobile device running Bluetooth inquiry

3. Bluetooth iBeacons
   Waveform details: three 376 µs transmissions 1ms apart at 30 msec interval
   REF_BCN setting: WURX_PHONE_ADV
   Transmitter: iOS device running the Atmosic app

For applications with a programmable wakeup transmitter that can generate custom OOK waveforms, please contact Atmosic Support for guidance on configuring the wakeup receiver.

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Reference Applications

The SDK supports two examples that use the wakeup receiver: WURX_adv and WURX_scan_adv. The following sections describe those examples in detail and compare their energy consumption. The examples use the wakeup receiver in a staged wakeup fashion, whereby a generic pattern is used to wake up all devices that are in range before Bluetooth LE is used to verify the Receiver. For Transmitters that can generate custom OOK waveforms, an application can use the wakeup receiver in a dedicated wakeup fashion, whereby a device-specific pattern is used to wake up just the Receiver. As a result, no Bluetooth LE verification is needed. Please contact Atmosic Support if you are interested in exploring such an application.

Example 1: WURX_adv (Global Wakeup)

In this example, the ATM2/3 is put in hibernation with the wakeup receiver enabled. Whenever a wakeup event is declared, the ATM2/3 transmits a series of beacons and then returns to hibernation. All of the beacon parameters are programmable via NVDS tags.
The Transmitter wakes up all devices that are in range, they all send beacons, and the Transmitter checks for the desired Receiver. Figure 3 shows how the system would work.

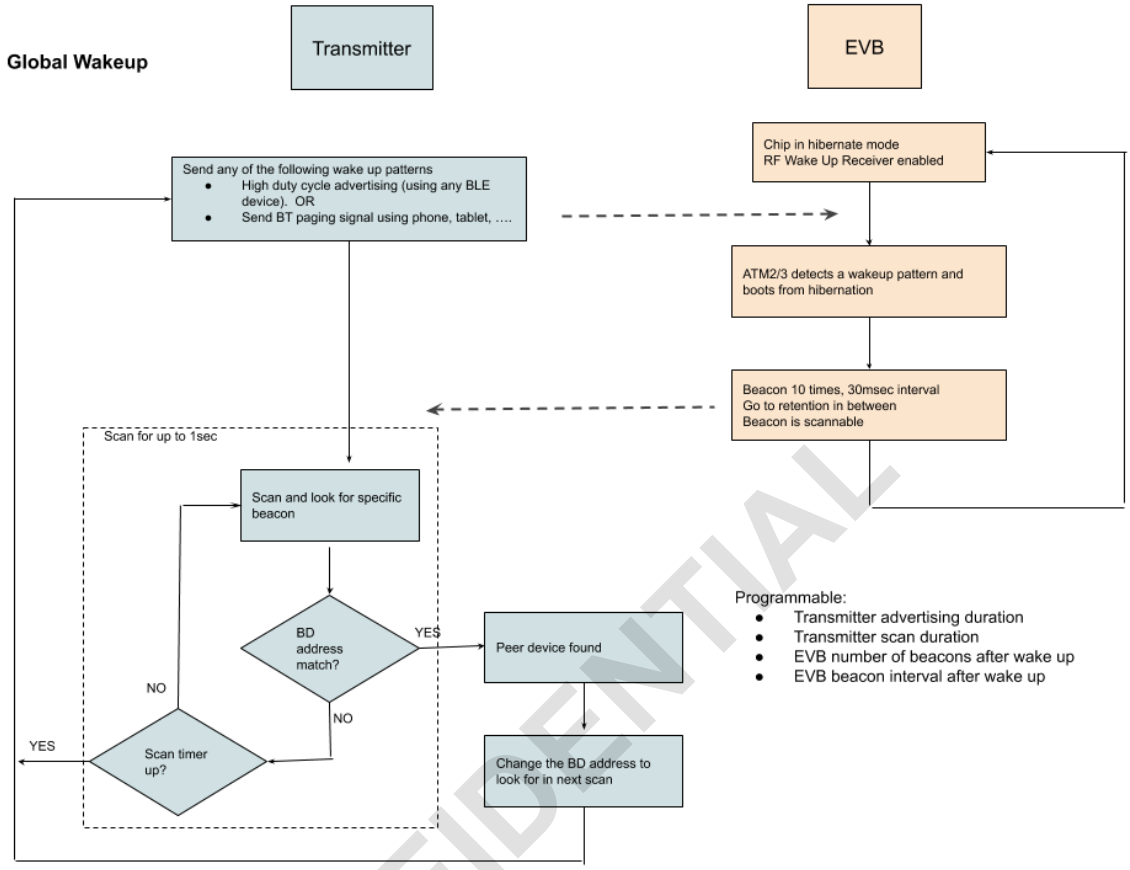


Figure 3 - Global Wakeup Flow Diagram

Figure 4 shows the VBAT current (in blue) and VDD1A current (in yellow) on an EVB during a wakeup event. The default NVDS tags configure the ATM2/3 to send 10 beacons (on 3 channels each) at a 30 ms interval.

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Figure 4 - Global Wakeup Current Consumption

Example 2: WURX_scan_adv (Targeted Wakeup)

This example only works with Transmitters that can send Bluetooth LE high duty cycle directed advertisements. The ATM2/3 is put in hibernation with the wakeup receiver enabled. Whenever a wakeup event is declared, the ATM2/3 scans for directed advertisements that match its Bluetooth Device (BD) address. If found, the ATM2/3 transmits a series of beacons and then returns to hibernation. All of the beacon parameters are programmable via NVDS tags. If no match is found, the ATM2/3 returns to hibernation without beaconing. As a result, the Transmitter wakes up all devices that are in range but only one sends beacons.

This example also tracks the number of consecutive false wakeups. If the number reaches a programmable threshold, the wakeup receiver is disabled and the ATM2/3 is configured to wake up at a programmable time interval. The ATM2/3 stays in this mode for a programmable period of time or until there is a valid targeted wakeup. Figure 5 shows how the system would work.

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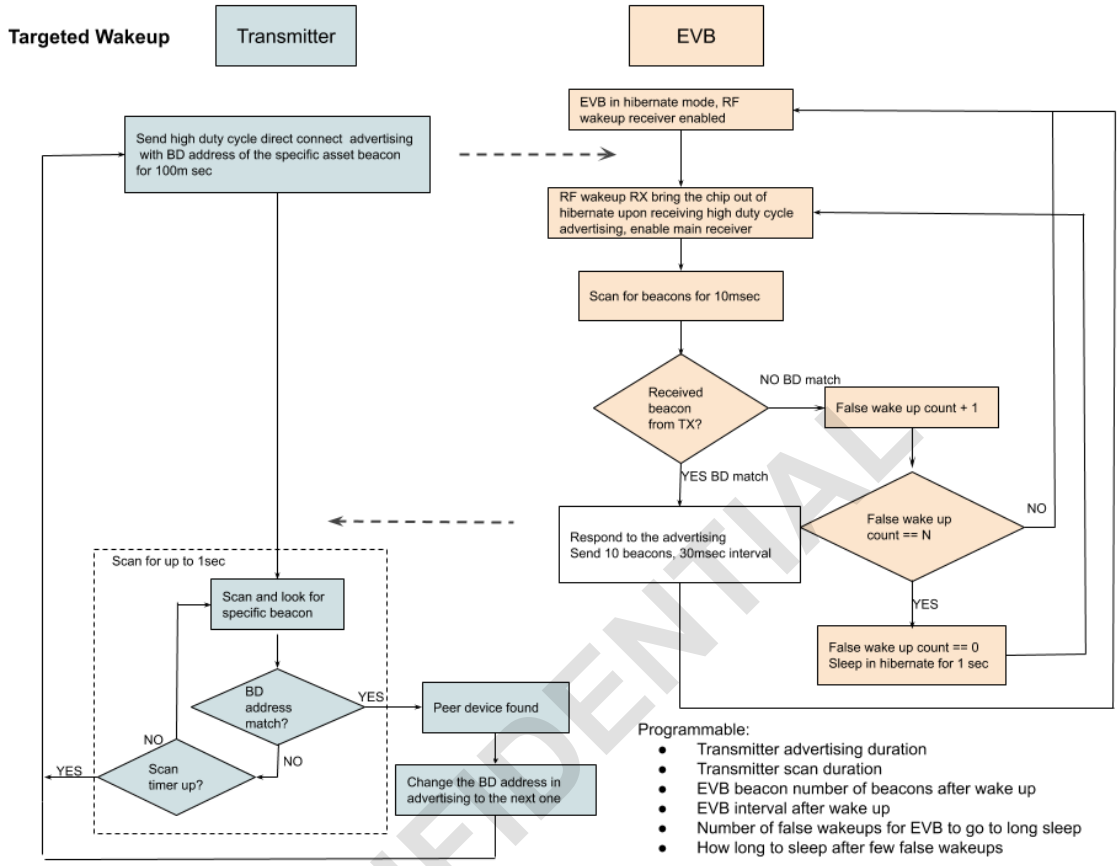


Figure 5 - Targeted Wakeup Flow Diagram

Figure 6 shows the VBAT current (in blue) and VDD1A current (in yellow) on an EVB during a valid wakeup event. The default NVDS tags configure the ATM2/3 to send 10 beacons (on 3 channels each) at a 30 ms interval.

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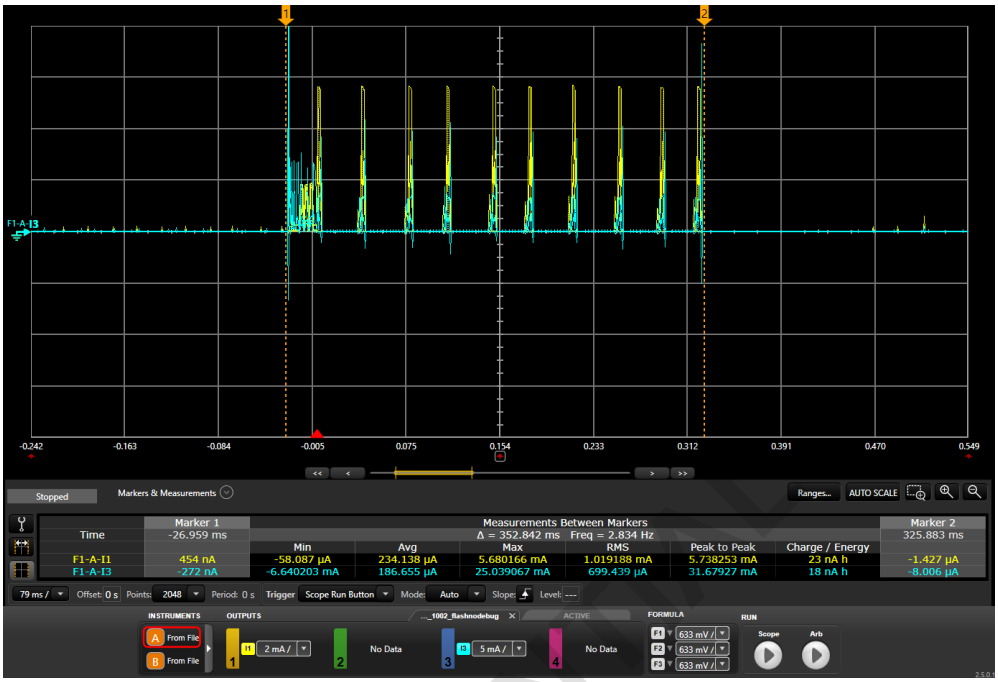


Figure 6 - Valid Targeted Wakeup Current Consumption

Figure 7 shows the VBAT current (in blue) and VDD1A current (in yellow) on an EVB during a false wakeup.



Figure 7 - False Targeted Wakeup Current Consumption

Energy Consumption Comparison

For global wakeup, every wakeup consumes ~157 µJ. For targeted wakeup, each valid wakeup consumes ~198 uJ while each false wakeup consumes ~60 µJ. As a result, targeted wakeup is recommended for applications that can experience a large number of false wakeups (e.g. noisy RF environment, high density of wakeup-enabled devices, etc).

Table 1 below compares the energy consumption of global and targeted wakeup with a non-wakeup beacon application with 100 ms interval. The results show significant savings for applications that beacon only when needed.

Table 1 - Energy Consumption Comparison Between Global and Targeted Wakeup and Non-wakeup Applications

	Global Wakeup	Targeted Wakeup	Non-wakeup Beacon
Beacons or valid wakeups per day	24	24	864,000
Energy consumed per beacon or valid wakeup	~157µJ	~198 µJ	~13 µJ
False wakeups per day	1,440	1,440	N/A
Energy consumed per false wakeup	~157µJ	~60 µJ	N/A
Sleep energy consumed per day	~220 mJ	~220 mJ	~670 mJ
Total energy consumed per day	**~450 mJ**	**~310 mJ**	~11,880 mJ

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Support for Two Wakeup Patterns

Running the WURX_adv example with REF_BCN set to WURX_PHONE_ADV enables support for two wakeup patterns: Bluetooth inquiry for Android devices and Bluetooth iBeacons for iOS devices. The ATM2/3 is put in retention with wakeup receiver enabled, and it switches between two sets of wakeup receiver configurations every second. The switch interval should be adjusted depending on the application's wakeup latency requirement. For switch intervals of 2 seconds or longer, hibernation would be more power efficient than retention.



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